IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 15 January 2026, accepted 31 January 2026, date of publication 6 February 2026, date of current version 11 February 2026.

*Digital Object Identifier 10.1109/ACCESS.2026.3661573*

RESEARCH ARTICLE

# AI-Powered Career Guidance: A Scalable Model for Personalized Recommendations

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**ABSTRACT** The primary objective of this study is to enhance career guidance in higher education institutions (HEIs) by integrating Holland's RIASEC model with advanced machine learning algorithms. Traditional career counselling methods, which often rely on matching interests with vocational categories, struggle to accommodate the complexities of the modern job market. This works aims to bridge that gap by utilizing data-driven techniques to provide more accurate, personalized, and scalable career recommendations. By aligning students' interests with real-world employability factors, the study seeks to improve the effectiveness of career decision-making processes in HEIs. The study uses the RIASEC model to identify student's professional interests and uses three well-known machine learning algorithms: Decision Trees, XGBoost, and Support Vector Machine (SVM) to predict students's; career pathways. To improve interpretability and reduce categorization mistakes, the models were further refined, making the results easier for HEI administrators to comprehend. In terms of prediction accuracy, the XGBoost algorithm outperformed other machine learning models as well as conventional techniques. It performed better in suggesting career pathways that closely matched participants's interests, as seen by its greater hit rates and smaller Euclidean distances. Promising outcomes were also displayed by the SVM algorithm, notably an increase in prediction accuracy above traditional career advising models. Graphs and charts, among other visual aids, were utilized to make complex patterns in the data easier to comprehend and analyze for administrators. Overall, compared to conventional RIASEC-based models, the machine learning-enhanced framework offered more precise and useful career choices. Applying machine learning techniques might significantly increase the accuracy and customisation of career advising in the RIASEC model. HEIs may provide more accurate job suggestions by utilizing algorithms like SVM and XGBoost, which efficiently match student's interests with future employability. Furthermore, the study offers organizations helpful policy recommendations that support the use of data- driven methods in career counseling services. In a labor market that is changing quickly, our insights help students and educational institutions make more strong and informed career decisions.

**INDEX TERMS** Career guidance, career recommendation, higher education, machine learning, RIASEC model.

## I. INTRODUCTION

Choices are the most difficult decisions an individual makes in his/her life [1], [2], [3]. Choices that are made typically depend on interests, skills, employability, etc [4], [5]. Holland's theory (RIASEC) perfectly underlines the relationship between interests and the career choices made by an individual [6], [7], [8], [9]. Six types of vocational interests are mentioned in this theory: realistic, investigative, artistic, social, enterprising, and conventional.

RIASEC's theory which is shown in Figure 1 suggests that individuals' interests influence their career choices, leading to better performance and job satisfaction when these interests align [10]. National Education Policy, which is currently being implemented in different schools, colleges, and universities across India, also aims to bridge the gap between interests pursued by an individual and their employability [11]. Career counseling services are crucial

The associate editor coordinating the review of this manuscript and approving it for publication was Massimo Cafaro.

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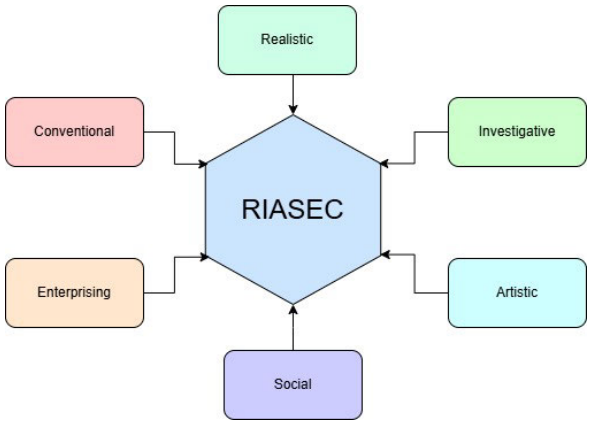



**FIGURE 1.** Hollandás RIASEC model with the descriptions of the six interest types.




**FIGURE 2.** Schematic steps involved in predictive analytics for the RIASEC model.

in colleges and universities to support students in making career choices and resolving decision making challenges, as individual career choices directly or indirectly impact a country's economic progress [12].

The primary research contributions of this study can be summarized as follows:

1) Integration of the traditional RIASEC framework with advanced machine learning techniques.
2) Development of accurate RIASEC-based career guidance using state-of-the-art machine learning models.
3) Enhancement of result interpretability through the use of graphical representations and visual analytics.
4) Addressing the challenge of sourcing accurate and relevant data for comprehensive analysis.
5) Leveraging machine learning to create a more adaptable system capable of handling the complexities of the modern workforce.
6) Conducting a rigorous evaluation of multiple machine learning algorithms to identify those with the highest relevance and accuracy.
7) Emphasizing the role of visual aids in simplifying complex relationships and patterns for improved understanding.

The study utilized supervised machine learning techniques like SVM, Decision Trees, and XGBoost to aid Higher Education Institutions (HEIs) in data analysis as shown in Figure 2. It also performed model optimization to reduce classification errors and improve model readability. The study also proposed policies to enhance educational system efficiency and student career support.

The major contributions of this research work are as follows:

1) A unique and interpretable machine learning framework is proposed to identify the key variables influencing students' career decisions.
2) A real-world career dataset consisting of 84,349 job aspirants was collected and analyzed based on RIASEC parameters.
3) The proposed framework's performance was evaluated using three widely used machine learning algorithms to validate the results and gain deeper insights into career decision patterns.
4) Based on the findings, a framework and policy recommendations were developed to assist higher education institutions (HEIs) and administrators in better understanding and responding to the evolving career landscape.

The O*NET profiler and other rule-based career tools are examples of traditional RIASEC-based systems that have long been the cornerstones of matching personal interests with professional paths. However, a number of significant constraints limit their usefulness. Because they assume constant and linear mappings between interest codes and work positions, these systems are fundamentally static and frequently fall short in capturing the nuances of individual variations or real-world career patterns. Such techniques provide theoretical framework, but they lack customization and adaptability, two attributes that are becoming more and more important in today's changing labor market, as previous research has shown [13], [14]. Machine learning (ML), on the other hand, offers a potent remedy for these issues. Large, diverse datasets may be used to train ML models, which can then identify multi-dimensional correlations and non-linear patterns that conventional approaches miss. Because of this flexibility, ML-based systems are able to provide more accurate, responsive, and customized job suggestions. The promise of machine learning approaches in career prediction tasks has been shown in recent research [14], [15], [16] which have shown considerable gains in prediction precision and user alignment. ML models like Support Vector Machine (SVM) and XGBoost fared noticeably better than conventional profile-matching techniques in the current study on a number of assessment measures. These methods outcomes shows the benefit of combining machine learning (ML) with the RIASEC technique to develop more sophisticated, data-driven, and user-focused career counseling solutions.

To assess the performance of the proposed career prediction models, we employed a set of metrics specifically tailored to the structure and objectives of RIASEC-based vocational guidance. These include:

1) Hit Rate: This metric assesses whether the predicted high-point RIASEC code aligns with the actual occupational high-point code. As a categorical accuracy measure, it directly supports the primary goal of recommending the most appropriate occupational category. Hit Rate is extensively used in career interest literature,

serving as an intuitive and interpretable benchmark for evaluating the effectiveness of career recommendation systems.

2) Euclidean Distance: This metric quantifies the numeric dissimilarity between the predicted and actual RIASEC profiles in a continuous six-dimensional space. It reflects how far the prediction deviates from the true interest pattern, offering a nuanced view of profile mismatch beyond just top-category accuracy.
3) Profile Correlation: By calculating the Pearson correlation between the predicted and actual RIASEC profiles, this metric evaluates the rank-order alignment of interest scores. It is particularly relevant in cases where individuals exhibit multiple strong interests across different RIASEC dimensions—a common scenario in real-world career decisions.

Together, these three metrics provide a complementary and multi-faceted evaluation:

1) Hit rate captures categorical agreement,
2) Euclidean distance measures numerical accuracy,
3) Profile correlation assesses structural similarity.

## II. LITERATURE REVIEW

A condensed version of the Big Five personality inventory, the BFI-10 was first presented by Rammstedt (2007) and is intended for rapid evaluations. Compared to the complete form, it loses substance, yet it is helpful when time is of the essence [13]. Nie et al. used behavioral indications from 4634 students using a variety of machine learning approaches in 2020 to forecast career preferences; nevertheless, they pointed out that a wider range of qualities was required. Although they did not include behavioral data, which limited its practical usefulness [14]. Levin et al. used Confirmatory Factor Analysis (CFA) to verify the Career Decision-Making Difficulties Questionnaire (CDDQ). Using the Five Factor Model [17]. Murwani et al. investigated high school career preferences; they found patterns but no predictive mechanism. Çağataylı and Çelebi used Random Forest classifiers and the Big Five personality characteristics to predict academic performance, although their feature set was limited. Li and Wang offer an optimal model for career planning that matches student characteristics with entrepreneurial positions using reliable neural networks and emphasizing interpretability and creativity [18]. Wang et al. presented a CBBOA SVM hybrid technique that incorporates Self Determination Theory. Interpretable machine learning (ML) (XGBoost, SHAP) was used on 18,000 student records by Yuan Wang et al.; however, they were missing certain personal characteristics. Although ambiguous questionnaire items and unbalanced data were observed, Song et al. employed RIASEC scales and several machine learning models to improve prediction accuracy [15]. Using a unique BPT + TF-IGM technique, Kamalesh and Bharathi identified the Big Five features, providing insights into personality but not directly relating them to a job [16]. Deep and discrete collaborative filtering models have been used in recent developments in educational data mining and explainable recommendation systems to improve interpretability and efficiency. In IEEE Transactions on Knowledge and Data Engineering, Zhu et al. introduced the Explainable Discrete Collaborative Filtering (EDCF) model as a solution to this problem [19]. Zhang et al. introduced a Cognitive Representation Optimization (CRO) framework to improve question embeddings in knowledge tracing models and improve the prediction of student learning behaviors and mastery level. AI-based employment advice is closely associated with the goal of predicting outcomes by linking cognitive, behavioral, and psychometric qualities [20]. Khlobystova et al. investigated user interests using statistical analysis and social media topic trends to derive RIASEC ratings [21]. The Counterfactual route-based Explainable Recommendation (CPER) framework was introduced by Li et al. [22]. This framework improves explanation integrity and confidence by overcoming the drawbacks of attention-based models, which makes it perfect for situations where transparency is crucial, such career and educational advice.The Explainable AI-based DSS with a web interface for real-time feedback was most recently created by Hossain et al. It performs well and is accessible, with an accuracy of 94.71 percent; nevertheless, more behavioral modeling is required [23]. Paul et al. used explainable AI techniques (SHAP and LIME) in conjunction with multiple machine learning models to investigate student dropout and academic performance [24]. They found that interpretable models, especially Logistic Regression, can effectively identify at-risk students and support early academic intervention in higher education institutions. In order to support data-driven academic monitoring and intervention strategies, Paul et al. examined several machine learning models for predicting university students' CGPA and highlighted the significance of model interpretability in comprehending important academic and demographic factors influencing student performance [25]. In order to predict student academic performance (GPA), Kalita et al. proposed a Bi-LSTM-based deep learning framework with SHAP-based interpretability [26]. This framework demonstrated better accuracy and statistical robustness than traditional machine learning models while staying focused on academic outcome prediction rather than career guidance or vocational alignment. Alonso et al. introduced a transformer-based framework that outperforms conventional job recommendation baselines in recruitment tasks by matching resumes with job descriptions and recommending pertinent abilities using O*NET taxonomy and semantic similarity [27]. Sharma et al. proposed a bilingual AI-driven job search platform NaukariCraft, that uses large language models to provide ATS-aware resume analysis and skill-based job recommendations, helping reduce language barriers and job-search fatigue for novice candidates [28]. Zhao suggests a transformer-based approach because traditional career prediction models are unable to capture the intricate relationships between behavioral and educational features [29].

**TABLE 1.** A comparative analysis of ML and behavioral data-based personality and career prediction methods.

| Author | Techniques Used | Data Details | Advantages | Limitations |
|---|---|---|---|---|
| Rammstedt (2007) | BFI-10 Personality Scale | 10-item short version | Time-efficient personality test | Less precise than full BFI |
| Nie et al. (2020) | DT, SVM, RF, LR | 4634 students, 4 indicators | Practical behavioral prediction | Less precise than full BFI |
| Levin et al. (2020) | CFA | Standardized questionnaire | Validated CDDQ structure | More features needed |
| Murwani et al. (2020) | Five Factor Model | HS questionnaire | Insight into interests | No behavioral traits used |
| Çağataylı and Çelebi (2021) | RF, Bagging | 2571 students | Links Big Five to success | No prediction model |
| Z. Wang et al. (2022) | CBBOA, SVM, BPNSS | 557 students | Psychological input with ML | Small data; no DL |
| Y. Wang et al. (2022) | XGBoost, TPE, SHAP | 18,000 students | Interpretable prediction | – |
| Song et al. (2022) | NN, KNN, RF, Elastic Net + RIASEC | 7866 students | Strong accuracy via personality | No personal traits used |
| Kamalesh and Bharathi (2022) | BPT + TFIGM (Text Mining) | Social media posts | Predicts Big Five traits | Vague inputs; class imbalance |
| Khlobystova et al. (2023) | Chi-sq, KW, Bayesian | SM subscriptions | Novel behavioral inference | Not tied to careers |
| Khlobystova et al. (2023) | Statistical Behavioral Inference | Topic-wise subscriptions | Scalable personality estimation | No psychometrics; privacy concerns |
| Hossain et al. (2025) | Explainable AI + ML Classifier | CS/SWE Google Form | 94.71% accuracy; web DSS | Needs validation/security |

A Comparative Analysis of ML and Behavioral Data-Based Personality and Career Prediction Methods is shown in Table 1.

**Research Gaps and Motivation:** Despite substantial progress in AI-powered career guidance, the literature reviewed in this study reveals several systematic and unresolved research gaps:

1) **Restricted Utilization of Extensive, Psychometrically Based Datasets:** Most existing studies rely on indirect behavioral proxies such as social media activity or small, institution-specific samples, typically comprising fewer than 10,000 records. The absence of large-scale, demographically diverse, and psychometrically validated datasets limits the robustness and generalizability of current career prediction models. Although standardized RIASEC-based assessments exist, they are rarely integrated with large datasets in machine learning-driven career guidance research.
2) **Excessive Dependency on Generic Assessment Measures:** The majority of prior work evaluates model performance using generic machine learning metrics such as accuracy, precision, recall, and F1-score. These measures fail to capture profile-level alignment, interest similarity, and degrees of mismatch across the six RIASEC dimensions, thereby overlooking the nuanced concept of vocational congruence central to Holland's theory.
3) **Insufficient Attention to Interpretability in Counseling Settings:** Many contemporary approaches prioritize deep learning or proprietary opaque systems that, while potentially improving predictive accuracy, lack transparency. Such opacity reduces their applicability in educational and counseling environments, where explainability, trust, and accountability are essential for informed decision-making by students and higher education administrators.
4) **Static and Non-Adaptive Career Recommendation Frameworks:** Both traditional RIASEC-based systems and several machine learning-enhanced models rely on static mappings between interests and occupations. These approaches inadequately model non-linear interest patterns, evolving career landscapes, and multi-interest profiles that increasingly characterize real-world vocational behavior.
5) **Disjointed Integration of Machine Learning and Psychometric Theory:** Although some studies incorporate personality traits or interest inventories, psychometric theory and machine learning are often treated as independent components. The current literature lacks a cohesive, theory-driven machine learning framework that preserves psychological validity while simultaneously enhancing predictive performance.

The proposed methodology motivates the development of a scalable ML framework for career guidance by identying these gaps.

## III. COMPARISON WITH EXISTING AI POWERED CAREER GUIDANCE SYSTEMS

To Distinguish the novelty of this proposed framework a comparative analysis with the existing AI based career counselling systems are presented along with the traditional O*NET baseline. This dedicated section includes the following domain of Skill graph based behavirol AI and psychometric career guidance systems: A. Pymetrics B. LinkedIn Career Explorer C. Eightfold.ai or Enterprise Talent Intelligence D. O*NET system (traditional)

**Pymetrics**: Pymetrics matches users with occupations based on their behavioral profiles by using machine learning to evaluate cognitive and emotional attributes through games inspired by neuroscience. The proprietary model raises

questions regarding transparency and validation versus psychometric norms, notwithstanding its efficacy in nonverbal tests. The importance of online presence on sites like LinkedIn and GitHub, AI for resume optimization, specialized job platforms, networking tactics, virtual interview preparation, and personal branding to help professionals adjust to the shifting job market are all covered in this chapter, which also looks at how digital tools improve job searches [30]. Pymetrics Career-Relevant Features combines Cognitive Traits, Emotional Traits, Behavioral Traits (from game-based neuro-assessments), Learnability Indicators (from Pymetrics–BCG analysis) to correlate person's neuro trait profiles with career role embeddings. **LinkedIn Career Explorer**: The dominant perspectives on the employability of graduates frequently place an emphasis on human capital while ignoring contextual and dispositional elements such as professional community connections and career identity. This chapter evaluates the use of LinkedIn for goal-setting and reflective employability reports [31] A common perception of employability as only human capital with limited connectivity capacities was found through qualitative research and student input. A linkedIn as a career counsellor framework consists of different phases like career path prediction, Identification of skill gap, career benchmarking, market trend awareness and networking recommendation [32].

**Eightfold.ai or Enterprise Talent Intelligence**: To integrate corporate personnel analytics, AI-based prediction, and career counseling tools to build a data-driven ecosystem that supports workforce planning, employee development, and institutional alignment between education and employment [33]. This recommendation system includes data integration layer, talent profiling layer, predictive intelligence layer, counselling layer followed by analytical and modelling phase results a career path [34].

**O*NET system**: A Self-reported RIASEC ratings are used by O*NET's Interest Profiler to recommend appropriate careers using tools which are theoretically solid and rule-based, lacks in behavioral customization and adaptive learning and instead rely on static mappings. This is not a sophisticated recommendation system, but they do function as a psychometric standard.

Table 2 shows comparison among several existing systems:

The proposed **RIASEC-based Machine Learning (ML) framework** demonstrates several distinctive advantages over existing AI-powered career guidance systems:

(i) **Psychometric Grounding:** The framework is trained and validated on a large-scale psychometric dataset comprising **81,000 participant profiles**, ensuring statistical robustness and behavioral diversity across demographic groups.

(ii) **RIASEC-Specific Evaluation Metrics:** Unlike generic accuracy-based models, the system employs **RIASEC-specific evaluation criteria** such as *Hit Rate*, *Euclidean Distance*, and *Profile Correlation*, capturing the semantic and psychological alignment between user profiles and occupational categories.

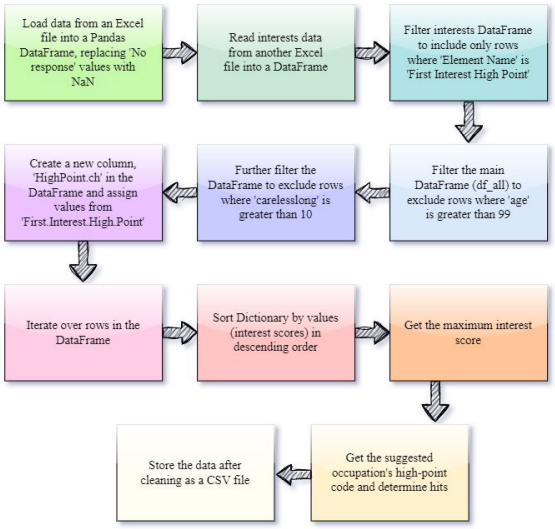



**FIGURE 3.** The steps to clean and pre-process the dataset.

(iii) **Interpretable Hybrid Modelling:** It integrates **interpretable ML algorithms**—Decision Trees, Support Vector Machines (SVM), Elastic Net, and XGBoost—optimized with RIASEC factor inputs to provide transparent, explainable recommendations suitable for professional counselling contexts.

(iv) **Human-Centric Design:** The framework supports **decision-making for students, career counsellors, and Higher Education Institutions (HEIs)** by combining data-driven insights with human expertise, ensuring that AI assists rather than replaces counsellor judgment.

## IV. DATA COLLECTION

Sources of data: O*NET, TIME Magazine online survey in 2016 Holland Code: Assign each individual a Holland Code based on their responses to the survey or assessment. Interest Scores: Determine people's interest in each of the six personality types. This study's dataset consists of 81,267 anonymised job seeker records that were gathered using standardized career guidance tests based on the O*NET and Holland Occupational Themes (RIASEC) frameworks. Job recommendations, interest indications, psychometric responses, and demographic information are all included in each record. Demographic data (age, gender, education, income, employment status, and country), answers to 20 career-related questions, standardized psychometric scales, interest scores based on the Holland Code, and suggested job roles with associated occupational codes are among the main feature categories. The dataset's demographic makeup is varied, with individuals spanning from 16 to 65 years of age, representing a range of genders and educational backgrounds from secondary school to doctorate degrees. Because the dataset contains people from other nations, it is more applicable in a wider range of geographical and cultural contexts. By guaranteeing that the data represents a diverse population, these characteristics

**TABLE 2.** Comparison of AI-powered career guidance systems.

| System | Primary Inputs | Approach | Adaptivity | Interpretability | Main Focus |
|---|---|---|---|---|---|
| **ONET** | Self-report RIASEC responses | Rule-based mapping | Static | High | General public / baseline profiler |
| **Pymetrics** | Game-based behavioral data | Proprietary ML models | Moderate | Medium | Job matching, employer tools |
| **LinkedIn Career Explorer** | User skills and job-market data | Graph / ML analytics | High | Low–Medium | Career transitions, skill gaps |
| **Eightfold.ai** | HR and enterprise skill data | Deep-learning talent intelligence | High | Low–Medium | Enterprise workforce solutions |
| **Proposed System** | RIASEC scores + demographic and occupational data | Interpretable ML models (SVM, DT, XGBoost, Elastic Net) | Moderate | High | Education and career counseling |

**TABLE 3.** An illustration of the identification of high-point codes and occupational RIASEC scores.

| Occupation | R | I | A | S | E | C | High-point Code |
|---|---|---|---|---|---|---|---|
| Learner 1 | 5.5 | 4.2 | 3.8 | 4.0 | 5.0 | 4.8 | R |
| Learner 2 | 3.5 | 6.0 | 4.0 | 3.8 | 4.2 | 6.0 | I, C |
| Learner 3 | 4.0 | 4.5 | 5.8 | 5.8 | 5.8 | 3.5 | A, S, E |

enhance the generalizability of prediction models derived from the data.

Credibility and dependability are enhanced by the fact that the dataset comes from reputable occupational databases and widely used career counseling approaches, particularly the O*NET occupational database for job mapping and the RIASEC model for interest profiling.

Algorithms Implementation: Comparing their accuracy.

The Figure 3 outlines a Python data processing workflow using Pandas DataFrames. Data is loaded from an Excel file, filtered, and filtered to retain rows with 'First Interest High Point'. The main DataFrame, 'HighPoint.ch,' is created, and the process iterates over rows, sorts interest scores, determines the high-point code of the participants and stores the cleaned data as a CSV file.

## V. METHOD

This analysis code and related research results are accessible via the Open Science Framework project website: https://osf.io/27rbh/.

### A. MEASURES

#### 1) VOCATIONAL INTERESTS: INTERESTS RELATED TO VOCATION

Participants scored each of the 16 items from the Personal Globe Inventory (PGI) Mini on a 7-point Likert scale to gauge their vocational interests. e. K-nearest neighbour imputation imputed missing data [35]. 2) HIGH-POINT CODE FOR OCCUPATION Participants were asked about their profession and aspirations using an interactive drop-down menu which is shown in Table 3. They chose their preferred career path or occupational membership. The occupational RIASEC scores and high-point codes for each job were obtained from the O*NET occupational interest database, demonstrating the representativeness of the occupation for each interest [15]. Sample: N = 84,349 people who answered an online survey 2 from TIME Magazine in 2016 made up our original sample [36]. The sample size was 81,267, with 62 average age of 38 years (SD = 13.46).

#### 2) METHODOLOGY AND ANALYTICAL APPROACH

To forecast an occupation's RIASEC scores that best fit a participant's assessed interests is the aim of the profile technique and the Machine learning based method that follows:

##### *a: PROFILE METHOD*

The profiling and ML-augmented methods aim to predict an occupation's RIASEC scores that align with a participant's interests. O*NET determines the link between RIASEC interest scores and participants' scores. The top five professions with the highest profile correlations are recommended for each participant. The first recommended job is chosen for forecasting [37].

The algorithm of this profile method is given in Algorithm 1:

##### *b: ML-AUGMENTED METHOD*

We used machine learning techniques that balance prediction accuracy, interpretability, and scalability in order to efficiently simulate the intricate correlations between RIASEC interest ratings and occupational outcomes. The models that were finally selected—Support Vector Machine (SVM), XGBoost, and Decision Tree—fit very nicely with the dataset's structure and the domain-specific needs of career counseling programs.

Because of its shown efficacy in high-dimensional spaces, such as those produced by 16-item PGI interest ratings, Support Vector Machine (SVM) was chosen. SVMs can predict non-linear connections without a high risk of overfitting because to the kernel technique. Previous studies have shown that SVM performs better on comparable prediction tasks, such as [12]. The gradient-boosted tree ensemble technique known as XGBoost is well known for its excellent accuracy when applied to structured tabular data. It generates well-calibrated predictions, effectively manages missing information, and has built-in regularization to avoid overfitting. Its use is further supported by its impressive

**Algorithm 1** RIASEC Profile Method for Career Prediction

**Input:** Participant RIASEC scores $P = \{P_R, P_I, P_A, P_S, P_E, P_C\}$
**Output:** Predicted career path (High-Point Code)

**Step 1: Load O*NET Data**
**for** each occupation $i$ from 1 to $N$ **do**
    Retrieve $J_i = \{J_{iR}, J_{iI}, J_{iA}, J_{iS}, J_{iE}, J_{iC}\}$
**end for**

**Step 2: Initialize**
Create empty list $C$ to store $(i, r_i)$

**Step 3: Compute Pearson Correlation**
Compute $\text{mean}_P = \frac{P_R+P_I+P_A+P_S+P_E+P_C}{6}$
**for** each occupation $i$ from 1 to $N$ **do**
    Compute $\text{mean}_{J_i} = \frac{J_{iR}+J_{iI}+J_{iA}+J_{iS}+J_{iE}+J_{iC}}{6}$
    Set `numerator` $\leftarrow 0$, `denom_P` $\leftarrow 0$, `denom_J` $\leftarrow 0$
    **for** each dimension $x \in \{R, I, A, S, E, C\}$ **do**
        $d_P \leftarrow P_x - \text{mean}_P$
        $d_J \leftarrow J_{ix} - \text{mean}_{J_i}$
        `numerator` $\leftarrow$ `numerator` $+ d_P \cdot d_J$
        `denom_P` $\leftarrow$ `denom_P` $+ d_P^2$
        `denom_J` $\leftarrow$ `denom_J` $+ d_J^2$
    **end for**
    **if** `denom_P` $\neq 0$ and `denom_J` $\neq 0$ **then**
        $r_i \leftarrow \frac{\texttt{numerator}}{\sqrt{\texttt{denom_P} \cdot \texttt{denom_J}}}$
    **else**
        $r_i \leftarrow 0$
    **end if**
    Append $(i, r_i)$ to list $C$
**end for**

**Step 4: Rank Occupations**
Sort $C$ in descending order by $r_i$

**Step 5: Recommend Top 5 Occupations**
Select top 5 occupations from $C \rightarrow$ Top5 = $[occ_1, \ldots, occ_5]$

**Step 6: Determine High-Point Code**
Let $BestMatch \leftarrow occ_1$
Extract high-point code from RIASEC profile of *BestMatch*

**Step 7: Output**
Print high-point code as predicted career path

**Step 8: Repeat**
Repeat steps 1 to 7 for each participant in the dataset

achievements in earlier interpretable ML frameworks, such as in Y. Wang et al [11].

**Algorithm 2** ML-Augmented Career Prediction Method

**Input:** Training data (RIASEC + PGI-Mini + careers), test data (interest ratings), actual RIASEC profiles
**Output:** Predicted RIASEC profiles, high-point codes, evaluation metrics (Hit Rate, Euclidean Distance, Profile Correlation)

**Preprocess Data:** Clean and normalize interest ratings
**Train ML Model:** Select algorithm (e.g., SVM, XGBoost), train with interest items as features, RIASEC as labels
**for** each participant in test data **do**
    Predict RIASEC profile $P = \{P_R, P_I, P_A, P_S, P_E, P_C\}$
    Extract high-point code from top-scoring dimension(s)
**end for**

**Evaluation: Hit Rate**
Initialize hit_count $\leftarrow 0$
**for** each participant **do**
    **if** predicted code == actual code **then**
        hit_count++
    **end if**
**end for**
Hit_Rate $\leftarrow$ hit_count / total participants

**Evaluation: Euclidean Distance**
**for** each participant **do**
    Compute $ED = \sqrt{\sum_i (P_i - A_i)^2}$
    Store distance
**end for**

**Evaluation: Profile Correlation**
**for** each participant **do**
    Compute Pearson correlation between predicted $P$ and actual $A$
    Store correlation
**end for**
**Output:** Predicted profiles, high-point codes, average metrics

Research already carried out using ML techniques like random forest, kNN and Elastic net [15]. The comparison among these ML techniques are shown in the results and discussion section.

The algorithm of ML augmented method is given in Algorithm 2:

Predicting career choices based on interest items filled out by participants. The machine learning models discover relationships between interest ratings and career decisions from a training sample. A fresh sample is provided to forecast ratings of occupational interest. The strategy uses the participants' PGI-Mini and RIASEC scores. Evaluation:

prediction accuracy The ML based methodology and profile method's prediction accuracy were evaluated using different parameters which measure the agreement between expected and actual career decisions [38].

High-Point Hit Rates High-point Hit Rate measures the agreement between the predicted high-point code (from the ML augmented or profile method) and the actual high point code. High-point Hit Rate = Number of Hits / Total Number of Participants Algorithm 1. Input: Predicted RIASEC codes, Actual RIASEC codes. 2. For each participant: Compare the predicted high-point code with the actual high-point- 3. If match, they Calculate the code. record Hit a hit.

Rate: Total hits / Total number of participants. 4. Output: High-point Hit Rate. Differential Hit Rate: Differential Hit Rate = High-point Hit Rate - Base Rate Where: Base Rate is the proportion of participants whose actual high-point code falls into a specific RIASEC category.

Euclidean distance is the difference between an individual's anticipated profession choice and their chosen RIASEC profile, while profile correlation is when job decisions and expected career choices coincide. Combining these three parameters provides a comprehensive analysis of prediction techniques. Table 4 contains some possible real-world scenarios along with the assessment measure that is best appropriate for each situation. Euclidean Distance quantifies the difference between the predicted RIASEC profile and the actual RIASEC profile across all six RIASEC dimensions.

$$\text{Euclidean Distance} = \sqrt{\sum (P_i - A_i)^2} \tag{1}$$

where: Pi is the predicted score for the ith RIASEC dimension. Ai is the actual score for the i-th RIASEC dimension. The sum is over all six RIASEC dimensions. A. Algorithm 1. Input: Predicted RIASEC scores P1, P2, …, P6, Actual RIASEC scores A1, A2, …, A6. 2. For each participant: Compute the squared difference for each RIASEC dimension: (Pi - Sum these - squared $(P_i - A_i)^2$. differences. Take the square root of the sum to get the Euclidean Distance. 3. Output: Euclidean Distance for each participant.

A "hit" is an individual's professional choice and predicted suggestion based on the traditional approach or Machine learning augmented technique. Table 5 shows the condition for both "hit," and "No hit." A hit occurs when the participant's high-point code coincides with the O*NET-recommended job profile. An overlapping point in the predictions using occupational high-point code is considered a "hit" if the participant's high-point code coincides with minimum tiered codes. The study analysed the high point hit rate of RIASEC, dividing the total number of hits by the sample size [38]. It also determined hit rates for jobless and employed individuals, and for each occupational group. The study evaluated prediction accuracy using Euclidean distance, comparing participants' reported and predicted interest scores. A lower distance indicates a more accurate prognosis [39], [40], [41]. Profile correlation evaluates

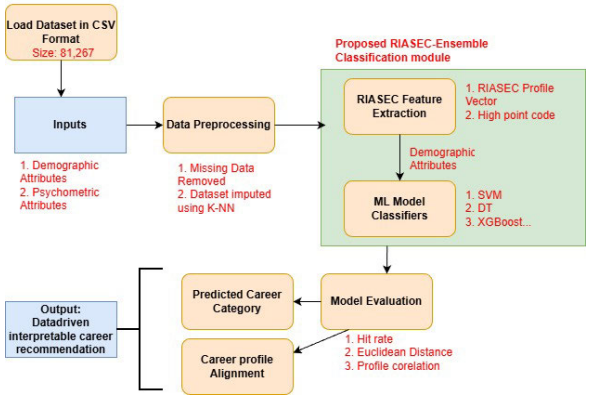



**FIGURE 4. Architecture of the proposed RIASEC–ensemble career prediction framework.**

prediction accuracy by ranking predicted and reported interest profiles [16]. Greater correlation indicates better forecast accuracy [42]. This study compares the efficacy of the ML augmented strategy with the profile method for professional choices. Profile Correlation measures the correlation between the predicted and actual RIASEC profiles, considering the rank order of scores across the six RIASEC dimensions.

$$\text{Profile Correlation} = \frac{\sum (P_i - \bar{P})(A_i - \bar{A})}{\sqrt{\sum (P_i - \bar{P})^2 \sum (A_i - \bar{A})^2}} \tag{2}$$

where: - Pi and Ai are the predicted and actual scores for the i-th RIASEC dimension, respectively. $\bar{P}$ and $\bar{A}$ are the mean of the predicted and actual scores on the six dimensions. Algorithm 1. Input: Predicted RIASEC scores P1, P2, …, P6, Actual RIASEC scores A1, A2, …, A6. 2. For each participant: Calculate the mean predicted score $\bar{P}$ and the mean actual score $\bar{A}$. Compute the covariance between the predicted and actual profiles.

Compute the variance for the predicted profile and the actual - Calculate the correlation profile. coefficient. 3. Output: Profile Correlation for each participant. This is shown in Figure 4.

Figure 4 depicts the data analysis pipeline, which involves gathering 81263 behavioral records from web sources, cleaning the data, creating features, producing RIASEC Scores, and analyzing High Point Code and Score. The data is then presented using violin plots, boxplots, and histograms, ensuring accurate distribution and relation to RIASEC scores. The process involves descriptive statistics, machine learning models like Decision Tree, SVM, and XGBoost, and performance evaluation using measures like Euclidean distance, hit rate, and profile correlation, ensuring a comprehensive data analysis from data gathering to assessment.

### B. CAREER-PATH SIMILARITY COMPUTATION

Career-path similarity in the proposed framework is computed within a psychometric feature space that represents each individual and occupation through their RIASEC profiles.

**TABLE 4.** Sample schemes for interest profiles with matching assessment metrics.

| S No. | Scheme | Example | Most Relevant Evaluation Metric |
|---|---|---|---|
| 1 | A person's interests in numerous RIASEC domains are similar. | Comparably high-interest rankings in I, A, S, and E parameters | Euclidean Distance |
| 2 | The top RIASEC domains are overlapped. | The maximum interest values are in the Artistic and Social domains, whereas the remaining RIASEC domains have significantly lower values. | Euclidean Distance |
| 3 | A person's RIASEC interest profile is significantly distinct. | Low-level interests in Social and Enterprising, high in Realistic, and intermediate in Investigative, Artistic, and Conventional. | High Point Hit Rate; Profile Correlation |
| 4 | An individual has significant opposition. | Extreme lack of interest in the Realistic domain. | Profile Correlation |
| 5 | someone has insufficient interests among all domains. | minimum level of interest in all RIASEC domains (R, I, A, E, and S) | Profile Correlation; High Point Hit Rate |

**TABLE 5.** Operationalization of the profile technique.

| O*NET recommended job profile | High-Point Code (O*NET) | High-Point Code (Self) | Hit |
|---|---|---|---|
| Electrician | R | E | No Hit |
| Preschool Teacher | S | S | Hit |

#### 1) RIASEC VECTOR REPRESENTATION

A six-dimensional vector representing RIASEC vocational interests is utilized to illustrate each participant's professional interest profile, with each component indicating the normalized score. Correspondingly, occupations from the ONET database are represented using RIASEC codes, also transformed into six-dimensional numerical vectors for comparison [43].

$$V = [R, I, A, S, E, C]$$

#### 2) SIMILARITY MEASURE

The directional alignment between a student's predicted RIASEC vector and an occupation's RIASEC profile is measured using cosine similarity, which determines how similar two career pathways are; values closer to 1 indicate higher similarity [44].

$$\text{Sim}(V_1, V_2) = \frac{V_1 \cdot V_2}{||V_1||\,||V_2||}$$

#### 3) ALTERNATIVE DISTANCE MEASURES FOR VALIDATION

In addition to cosine similarity, *Euclidean Distance (ED)* and *Profile Correlation (PC)* are used as supplementary metrics. Euclidean Distance measures the numerical deviation between two RIASEC profiles, while Profile Correlation assesses the rank-order relationship of interests based on Holland's theory of congruence. Together, these three measures provide a comprehensive understanding of similarity in both direction and magnitude.

#### 4) INTEGRATION WITH CAREER PREDICTION MODEL

The machine learning models (e.g., SVM, Decision Tree, and XGBoost) predict the most likely RIASEC profile for a user. The predicted profile is then compared with occupational profiles using the combined similarity metrics (cosine and Euclidean), and the top-ranked occupations are recommended as the most compatible career pathways.

## VI. RESULTS AND DISCUSSION

### A. IMPLEMENTATION DETAILS

A dataset of career interests from the Open Science Framework project was used in the training procedure. The machine used for all trials was a Windows 11 Home Single Language system (version 23H2, OS build 22631.5699) with an Intel®Core™i5-1235U 12th Gen CPU (1.30 GHz), Intel®Iris®Xe graphics, and 16 GB of RAM. A sample of 81,267 occurrences was used to train the model during 20 epochs.

### B. ABLATION STUDY AND MODEL VARIANT ANALYSIS

An ablation study was conducted to determine which components of the proposed RIASEC-based machine learning framework had the greatest impact on the accuracy and interpretability of career interest predictions. Important modules were removed or modified to observe variations in prediction performance [45].

#### 1) COMPONENTS CONSIDERED FOR ABLATION

The full framework consists of three major components:

(A) **Psychometric features** derived from the RIASEC scale and PGI-Mini items.
(B) **Extended demographic and academic attributes** (e.g., age, gender, stream, and GPA).
(C) **Machine learning model ensemble**, combining algorithms such as XGBoost, Decision Tree, and Elastic Net with RIASEC-specific evaluation metrics (Hit Rate, Euclidean Distance, and Profile Correlation).

#### 2) EXPERIMENTAL SETUP

Four model variants were tested on the same dataset (≈81,000 records) with identical train–test splits:

1) **Full Model (A + B + C)** — the complete proposed framework.
2) **Model without Demographic Features (A + C)** — isolates psychometric contribution.
3) **Model without Psychometric Inputs (B + C)** — tests reliance on non-psychometric predictors.
4) **Single-Model Variant (A + B)** — replaces the ensemble with a single Decision Tree to assess ensemble benefits.

**TABLE 6.** Ablation study and model variant analysis.

| Method | Hit Rate | Inference / Observation |
|---|---|---|
| **Full Model (A + B + C)** | 0.83 | Best overall performance; balanced accuracy and interpretability. |
| **Without Demographic Features (A + C)** | 0.79 | Slight drop; demographic features contribute secondary improvements. |
| **Without Psychometric Inputs (B + C)** | 0.67 | Major decline; psychometric features are most influential. |
| **Single-Model Variant (A + B)** | 0.75 | About 10% lower accuracy; ensemble improves robustness. |

**TABLE 7.** Five-fold cross-validation results.

| Fold | Loss | Accuracy (%) | Train + Test Time (s) |
|---|---|---|---|
| 1 | 0.338624 | 66.1376 | 1.548723 |
| 2 | 0.333026 | 66.6974 | 1.202485 |
| 3 | 0.332431 | 66.7569 | 1.239243 |
| 4 | 0.336061 | 66.3939 | 1.231565 |
| 5 | 0.342521 | 65.7479 | 1.265275 |
| **Average** | **0.336533** | **66.3467** | **1.297458** |

Psychometric elements are crucial for predicting accuracy; their removal led to the biggest performance reduction, with the average hit rate falling from 0.83 to 0.67 and profile correlation falling by almost 18 in percent. A lesser decline (hit rate 0.83 to 0.79) was seen when demographic factors were excluded, indicating that they have secondary value. Performance was reduced by around 10 in percent when using a single Decision Tree, highlighting the benefits of ensemble modeling. The whole model performed best, balancing generality and interpretability, proving that while demographic characteristics provide only modest gains in particular predictions, psychometric factors are crucial. The result of this ablation study is shown in the Table 6.

A five-fold cross-validation was used to assess the performance of the suggested model, yielding an average accuracy of 66.35 percent, a mean loss of 0.3365, and a computation time of 1.297 seconds per fold. The details are shown using the following Table 7:

The Hit Rate, Euclidean Distance, and Profile Correlation mean and standard deviation from five-fold cross-validation are shown in Table 8. With XGBoost displaying the lowest variance in distance-based accuracy and SVM displaying slightly larger variability in hit rate, all models' low standard deviations point to consistent and repeatable performance.

### C. RUNTIME

The suggested approach demonstrated computational efficiency and feasibility for large-scale deployment when implemented under the same experimental criteria as previously discussed, requiring an average training time of 0.7363 seconds and a testing time of 0.0230 seconds.

### D. DISCUSSION

The implementation of SVM, Decision Tree, XGBoost algorithm has been accomplished for this work. This has been accomplished in the following phases:

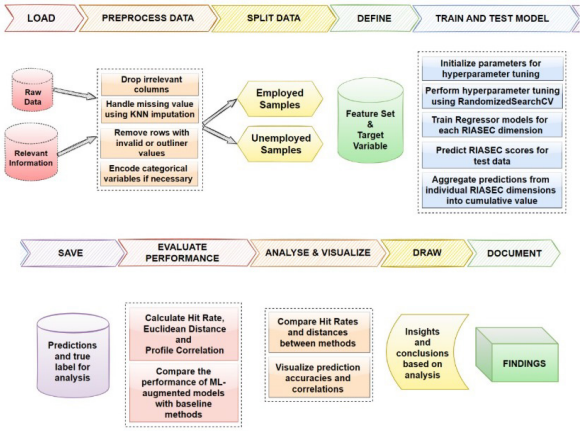



**FIGURE 5.** Methodology for applying and assessing machine learning models for RIASEC dimensions prediction.

In Figure 5, The implementation process involves loading a raw dataset, pre-processing it to remove irrelevant columns, and dividing it into employed and unemployed samples for targeted analysis. Hyperparameters are tuned using RandomizedSearchCV, and separate regression models are trained for each RIASEC dimension. Predictions are generated and saved for detailed performance analysis, with results compared to baseline methods and findings documented for comprehensive record-keeping.

Table 9 shows a comparative study of the performance metrics for the Support Vector Machine (SVM) and the Profile Method. Overall, SVM outperforms the Profile Method in hit rates and Euclidean distances, although the profile correlation results are mixed. SVM shows a higher correlation for OM (Occupational membership) but the same for VA (Vocational Aspirant).

Table 10 compares hit rates for RIASEC categories using different methods and conditions. The Profile Method improves in 'R' and 'A' categories, while SVM outperforms in 'E' with a significant increase and a higher hit rate. Then, the Profile Method and SVM are evaluated in VA conditions, with the Profile Method showing higher performance in the 'I' and 'C' categories, while SVM improves in the 'A' and 'E' categories, highlighting its effectiveness in predicting RIASEC dimensions.

Table 11 compares Euclidean distances across different RIASEC categories using two methods: OM-Profile and OM-SVM, as well as VA Profile and VA-SVM. In the Realistic category, OM-SVM shows lower distances than OM-Profile and VA-SVM. In the Investigative category, both methods have slightly lower distances than profile methods. In the Artistic category, distances are relatively close across methods. In the Enterprising category, OM-SVM and VA-SVM have significantly lower distances than profile methods.

Table 12 shows profile correlation comparisons across RIASEC categories using various methods. Realistic (R) shows slight positive correlations for OM-SVM and OM-ML

**TABLE 8.** Mean ± standard deviation of evaluation metrics (5-Fold Cross-Validation).

| Model | Hit Rate (Mean ± SD) | Euclidean Distance (Mean ± SD) | Profile Correlation (Mean ± SD) |
|---|---|---|---|
| Decision Tree | $0.28 \pm 0.02$ | $4.54 \pm 0.11$ | $0.32 \pm 0.03$ |
| SVM | $0.40 \pm 0.03$ | $4.24 \pm 0.09$ | $0.23 \pm 0.04$ |
| XGBoost | $0.30 \pm 0.02$ | $4.19 \pm 0.08$ | $0.40 \pm 0.03$ |

**TABLE 9.** Overall comparison among several parameters.

| Method | OM-HR | OM-Euc | OM-PC | VA-HR | VA-Euc | VA-PC |
|---|---|---|---|---|---|---|
| Profile Method | 0.27 | 5.19 | 0.23 | 0.34 | 4.99 | 0.28 |
| SVM | 0.40 | 4.24 | 0.43 | 0.31 | 4.56 | 0.28 |

**TABLE 10.** Hit Rate comparison as per each RIASEC category.

| RIASEC | OM-Base | OM-Profile | OM-SVM | OM-Prof Diff | OM-SVM Diff | VA-Base | VA-Profile | VA-SVM | VA-Prof Diff | VA-SVM Diff |
|---|---|---|---|---|---|---|---|---|---|---|
| R | 0.07 | 0.16 | 0.00 | 0.09 | -0.07 | 0.08 | 0.18 | 0.10 | 0.10 | 0.02 |
| I | 0.18 | 0.27 | 0.04 | 0.09 | -0.14 | 0.21 | 0.30 | 0.00 | 0.09 | -0.21 |
| A | 0.06 | 0.44 | 0.25 | 0.38 | 0.19 | 0.23 | 0.43 | 0.23 | 0.20 | 0.00 |
| S | 0.16 | 0.21 | 0.15 | 0.05 | -0.01 | 0.13 | 0.20 | 0.06 | 0.07 | -0.07 |
| E | 0.35 | 0.38 | 0.82 | 0.03 | 0.47 | 0.29 | 0.42 | 0.77 | 0.13 | 0.48 |
| C | 0.19 | 0.12 | 0.20 | -0.07 | 0.01 | 0.06 | 0.26 | 0.00 | 0.20 | -0.06 |

**TABLE 11.** Euclidean distance comparison as per Each RIASEC Category.

| RIASEC | OM-Profile | OM-SVM | VA-Profile | VA-SVM |
|---|---|---|---|---|
| R | 5.79 | 4.92 | 5.93 | 5.18 |
| I | 5.04 | 4.83 | 5.01 | 5.10 |
| A | 4.57 | 4.81 | 4.72 | 4.39 |
| S | 5.28 | 4.89 | 5.14 | 4.96 |
| E | 5.03 | 3.77 | 4.85 | 4.06 |
| C | 5.54 | 3.73 | 5.08 | 4.70 |

**TABLE 12.** Profile correlation comparison across RIASEC categories.

| RIASEC | OM-Profile | OM-SVM | VA-Profile | VA-SVM | OM-ML | VA-ML |
|---|---|---|---|---|---|---|
| R | 0.05 | 0.07 | 0.01 | 0.05 | 0.07 | -0.05 |
| I | 0.24 | 0.16 | 0.25 | 0.07 | 0.16 | -0.07 |
| A | 0.33 | 0.04 | 0.31 | 0.30 | -0.04 | 0.30 |
| S | 0.16 | 0.04 | 0.22 | 0.03 | 0.04 | 0.03 |
| E | 0.31 | 0.69 | 0.37 | 0.62 | 0.69 | 0.62 |
| C | 0.16 | 0.68 | 0.28 | 0.35 | 0.68 | 0.35 |

**TABLE 13.** Overall Comparison among different parameters.

| Method | OM Hit Rate | OM Euclidean | OM Profile Corr | VA Hit Rate | VA Euclidean | VA Profile Corr |
|---|---|---|---|---|---|---|
| Profile Method | 0.28 | 5.18 | 0.23 | 0.34 | 4.97 | 0.28 |
| Decision Tree | 0.28 | 4.54 | 0.32 | 0.31 | 4.78 | 0.24 |

Augmented, Investigative (I) shows moderate positive correlations for OM-Profile and VA-Profile, and Artistic (A) shows generally positive correlations. The study found that OM Profile and VA-Profile methods show modest positive correlations in the Social category, while other methods show lower or near-zero correlations. In the Enterprising category, strong positive correlations were observed for OM-SVM, VA-SVM, and VA-ML Augmented. SVM and ML-augmented methods generally show higher positive correlations.

*Decision Tree:* Table 13 compares all the parameters outcome between the traditional method and the Decision Tree ML technique for both OM and VA datasets. Both methods achieved a similar hit rate, but the Decision Tree method showed better accuracy in distance metrics and a stronger relationship between predicted and actual profiles. In the VA dataset, the Profile method had a slightly higher hit rate but better Euclidean distance and profile correlation accuracy.

Table 14 compares hit rates across RIASEC categories for OM and VA datasets using Base Rate, Profile Method, and Decision Tree approaches. The Decision Tree method has a higher hit rate in the Realistic category, while the Profile method outperforms it in the Investigative, Artistic, Social, Enterprising, and Conventional categories. However, the Profile method performs better in some categories.

Table 15 compares Euclidean distances across RIASEC categories for OM and VA datasets using the Profile and Decision Tree methods. The Decision Tree method achieved lower Euclidean distances in the Realistic category, Investigative category, Artistic category, Social category, Enterprising category, and Conventional category. The VA dataset slightly outperformed the Decision Tree method in some categories. The Decision Tree method showed better accuracy in most categories for the OM dataset, while the Profile method slightly outperformed in a few instances. The Decision Tree method generally exhibited lower Euclidean distances, indicating better accuracy, across most RIASEC categories.

Table 16 compares profile correlations across RIASEC categories using various methods. The OM Decision Tree method has low realistic correlations, high investigative and artistic correlations, moderate social and enterprising correlations, and the lowest conventional correlations. It has comparable overall forecast accuracy to conventional profiling techniques but performs better at shorter distances using Euclidean Distances (ED). However, it performs worse in Realistic, Investigative, Artistic, and Social categories, but outperforms conventional techniques in Enterprising and Conventional categories. Despite some lag in some areas, the Decision Tree provides more accurate results.

*XG BOOST:* Table 17 compares traditional profiling methods and XGBoost performance metrics. XGBoost outperforms traditional methods in hit rates, Euclidean distances, and profile correlations. It achieves lower distances for classification accuracy and higher correlations for profile correlation. Overall, XGBoost shows superior accuracy and predictive power compared to traditional methods.

Table 18 compares hit rates across RIASEC categories for convensional profiling methods and XGBoost. In the Realistic category, XGBoost's hit rate decreases compared to the traditional Profile Method. In the Investigative category, XGBoost decreases, while the Artistic category substantially increases. XGBoost slightly outperforms the Profile Method in the Social category, while the Enterprising category shows

**TABLE 14.** Hit rate comparison as per each RIASEC Category.

| RIASEC | OM Base | OM Profile | OM Tree | OM Pro Diff | OM Tree Diff | VA Base | VA Profile | VA Tree | VA Pro Diff | VA Tree Diff |
|---|---|---|---|---|---|---|---|---|---|---|
| R | 0.07 | 0.18 | 0.29 | 0.11 | 0.22 | 0.08 | 0.18 | 0.15 | 0.10 | 0.07 |
| I | 0.17 | 0.26 | 0.12 | 0.09 | 0.05 | 0.21 | 0.30 | 0.32 | 0.09 | 0.11 |
| A | 0.06 | 0.45 | 0.17 | 0.39 | 0.11 | 0.23 | 0.42 | 0.20 | 0.19 | 0.03 |
| S | 0.16 | 0.20 | 0.34 | 0.04 | 0.18 | 0.12 | 0.23 | 0.15 | 0.11 | 0.03 |
| E | 0.35 | 0.38 | 0.38 | 0.03 | 0.03 | 0.29 | 0.42 | 0.55 | 0.13 | 0.26 |
| C | 0.19 | 0.13 | 0.22 | 0.06 | 0.03 | 0.07 | 0.20 | 0.00 | 0.13 | 0.07 |

**TABLE 15.** Euclidean distance comparison as per each RIASEC category.

| RIASEC | OM-Profile | OM-Tree | VA-Profile | VA-Tree |
|---|---|---|---|---|
| R | 5.79 | 5.15 | 5.80 | 5.85 |
| I | 5.07 | 5.46 | 5.00 | 4.91 |
| A | 4.52 | 4.92 | 4.76 | 4.66 |
| S | 5.29 | 4.53 | 5.06 | 5.04 |
| E | 5.00 | 4.05 | 4.81 | 4.35 |
| C | 5.49 | 4.32 | 5.08 | 5.22 |

**TABLE 16.** Profile correlation comparison as per each RIASEC category.

| RIASEC | OM-Profile | OM-Tree | VA-Profile | VA-Tree |
|---|---|---|---|---|
| R | 0.04 | 0.14 | 0.02 | 0.01 |
| I | 0.24 | 0.05 | 0.24 | 0.19 |
| A | 0.34 | 0.14 | 0.31 | 0.27 |
| S | 0.16 | 0.21 | 0.23 | 0.16 |
| E | 0.32 | 0.49 | 0.38 | 0.35 |
| C | 0.18 | 0.42 | 0.28 | 0.15 |

**TABLE 17.** Overall Comparison of Hit Rates, Euclidean Distance, and Profile Correlation.

| Method | OM Hit Rate | OM Euc. | OM Prof. Corr. | VA Hit Rate | VA Euc. | VA Prof. Corr. |
|---|---|---|---|---|---|---|
| Profile Method | 0.27 | 5.19 | 0.23 | 0.34 | 4.99 | 0.28 |
| XGBoost | 0.30 | 4.19 | 0.40 | 0.35 | 4.35 | 0.33 |

the strongest performance. XGBoost performs better in the Enterprising and Conventional categories.

Table 19 compares Euclidean Distance across various RIASEC categories using the Profile Method and XGBoost. In the Realistic category, XGBoost outperforms the Profile Method in terms of accuracy. In the Investigative category, XGBoost outperforms the Profile Method in terms of accuracy. In the Artistic category, the Profile Method slightly outperforms XGBoost. Overall, XGBoost generally shows better accuracy.

Table 20 compares the correlations between the Profile Method and XGBoost in different RIASEC categories, focusing on OM and VA. In the R category, the Profile Method shows slight positive correlations, while XGBoost shows negative correlations for OM and VA. Both methods show positive correlations in the I category, while XGBoost has lower positive correlations. Both methods show stronger positive correlations in the A category, while XGBoost has very high positive correlations. The results show that The XGBoost algorithm outperforms conventional profiling techniques in several key areas, including prediction accuracy and hit rate. It also shows reduced Euclidean Distances, indicating higher prediction accuracy. However, it performs inconsistently in RIASEC categories, particularly in the R, I, A, and S categories. However, it outperforms the profile method in the E and C categories. Despite underperformance in some areas, XGBoost performs as well or better than conventional profiling in profile correlation, indicating its potential as a strong substitute for conventional profiling.

Table 21 compares the machine learning techniques based on HIT Rate, Euclidean Distance, and Profile Correlation metrics to predict occupational membership categories [15]. SVM has the best HIT Rate, but its profile correlation is weaker, and its Euclidean Distance is higher. Although they have a modest profile correlation, decision trees have the lowest HIT rate. XGBoost has a somewhat better HIT Rate but excels in Profile Correlation. Random Forest and Elastic Net exhibit the greatest profile correlation.

Table 22 compares different algorithms indicates different performance levels in terms of Euclidean Distance, Profile Correlation, and Hit Rate [15]. The study found that Elastic Net, Random Forest, and Neural Network were the most effective networks, with the highest hit rate and profile correlation. XGBoost also showed competitiveness, with a similar rate. SVM and Decision Tree algorithms performed worse, with smaller profile correlations and lower hit rates. Elastic Net, Random Forest, and Neural Network are recommended for their superior accuracy, profile alignment, and predictive performance.

By combining machine learning (ML) with a sizable, interest-based dataset, this work considerably improves career recommendation approaches, exceeding previous ML implementations as well as conventional RIASEC-matching systems.

Static look-up mappings between high-point interest codes and pre-established occupational categories are usually how traditional rule-based RIASEC systems work. These systems produce generic and frequently out-of-date career recommendations since they lack the ability to learn dynamically or customize. Similarly, human counselors often rely on subjective assessment and restricted psychometric inputs, which might introduce bias and variability, even when they are useful in interpretative and emotional situations. Contrary our machine learning enhanced method yields significant gains:

**TABLE 18.** Hit Rate Comparison as per Each RIASEC Category.

| RIASEC | OM-Base | OM-Profile | OM-XGBoost | OM-Pro Diff | OM-XGB Diff | VA-Base | VA-Profile | VA-XGBoost | VA-Pro Diff | VA-XGB Diff |
|---|---|---|---|---|---|---|---|---|---|---|
| R | 0.07 | 0.16 | 0.05 | 0.09 | -0.02 | 0.08 | 0.18 | 0.00 | 0.10 | -0.08 |
| I | 0.18 | 0.27 | 0.00 | 0.09 | -0.18 | 0.21 | 0.30 | 0.34 | 0.09 | 0.13 |
| A | 0.06 | 0.44 | 0.00 | 0.38 | -0.06 | 0.23 | 0.43 | 0.01 | 0.20 | -0.22 |
| S | 0.16 | 0.21 | 0.24 | 0.05 | 0.08 | 0.13 | 0.20 | 0.20 | 0.07 | 0.07 |
| E | 0.35 | 0.38 | 0.51 | 0.03 | 0.16 | 0.29 | 0.42 | 0.85 | 0.13 | 0.56 |
| C | 0.19 | 0.12 | 0.35 | -0.07 | 0.16 | 0.06 | 0.26 | 0.12 | 0.20 | 0.06 |

**TABLE 19.** Euclidean distance comparison as per Each RIASEC category.

| RIASEC | OM - Profile Method | OM - XGBoost | VA - Profile Method | VA - XGBoost |
|---|---|---|---|---|
| R | 5.79 | 5.15 | 5.93 | 5.71 |
| I | 5.04 | 4.87 | 5.01 | 4.60 |
| A | 4.57 | 4.97 | 4.72 | 4.35 |
| S | 5.28 | 4.39 | 5.14 | 4.66 |
| E | 5.03 | 3.69 | 4.85 | 3.81 |
| C | 5.54 | 3.91 | 5.08 | 4.34 |

**TABLE 20.** Profile correlation comparison as per each RIASEC ategory.

| RIASEC | OM-Profile | OM-XGBoost | VA-Profile | VA-XGBoost |
|---|---|---|---|---|
| R | 0.05 | -0.07 | 0.01 | 0.01 |
| I | 0.24 | 0.06 | 0.25 | -0.33 |
| A | 0.33 | -0.08 | 0.31 | 0.33 |
| S | 0.16 | 0.15 | 0.22 | 0.04 |
| E | 0.31 | 0.70 | 0.37 | 0.68 |
| C | 0.16 | 0.56 | 0.28 | 0.44 |

**TABLE 21.** Occupational membership category prediction results.

| Algorithms | Hit Rate | Euclidean Distance | Profile Correlation |
|---|---|---|---|
| SVM | 0.40 | 4.24 | 0.23 |
| Decision Tree | 0.28 | 4.54 | 0.32 |
| XGBoost | 0.30 | 4.19 | 0.40 |
| Elastic Net | 0.34 | 4.07 | 0.44 |
| Random Forest | 0.33 | 4.08 | 0.44 |
| Neural Network | 0.34 | 4.11 | 0.43 |
| K-NN | 0.34 | 4.11 | 0.43 |

**TABLE 22.** Vocational aspiration category prediction results.

| Algorithms | Hit Rate | Euclidean Distance | Profile Correlation |
|---|---|---|---|
| SVM | 0.34 | 4.99 | 0.28 |
| Decision Tree | 0.31 | 4.78 | 0.24 |
| XGBoost | 0.35 | 4.35 | 0.33 |
| Elastic Net | 0.36 | 4.26 | 0.35 |
| Random Forest | 0.36 | 4.28 | 0.34 |
| Neural Network | 0.36 | 4.28 | 0.35 |
| K-NN | 0.34 | 4.32 | 0.33 |

The profile matcher's success rate of 0.27 was much lower than the 0.40 attained by Support Vector Machine (SVM) (see Table 5 ).

While the profile technique only produced profile correlation scores of 0.23, Elastic Net and Random Forest produced values of 0.43, suggesting a more accurate representation of users' interest hierarchies.

The tighter alignment between the anticipated and real RIASEC profiles was confirmed by XGBoost's reduced Euclidean distance (4.19), which was lower than that of conventional techniques.

Additionally, this work is notable for its scalability and performance in comparison to other ML-based systems. Song et al for instance, used a neural network to a dataset of 7,866 people, but they were unable to get the same degree of profile alignment or hit rate [15]. Interpretable models for career prediction were used by Wang et al. [11], but they used a smaller and less demographically diverse sample. Furthermore, an explainable AI system for student career counseling was presented by Hossain et al. [23], but our method improves generalizability by applying comparable concepts to a far bigger and more varied dataset of over 80,000 people. Numerous earlier methods lacked verified behavioral interest ratings [17], were centered on social media trends [21], or were restricted to personality trait modeling [1]. This study provides a scalable, data-driven, and interpretable framework for contemporary vocational coaching by fusing a strong psychometric basis (the PGI Mini), thorough occupational mapping using O*NET, and high-performing machine learning models.

**Ethical and Bias Considerations:** A responsible and transparent AI based career counselling system must be designed to find the suitable career as it's a lifetime decision. Balanced sampling and subgroup evaluation can help in protecting data privacy. This system taken care of all important factors which may seek to provide fair and scalable career counsling.

## VII. CONCLUSION

To sum up, this research investigated how machine learning algorithms could improve how people are classified into the RIASEC job categories: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional. The study aimed to improve the classification of people into RIASEC job categories using machine learning algorithms. The results showed that Elastic Net was the most efficient approach, with better accuracy and alignment with individual and total label distributions. Elastic Net performed well when combined with other algorithms like SVM and Random Forest.

Several limitations must be noted, even if this work offers insightful information on improving RIASEC-based job counseling using machine learning. First, the dataset is dated (collected in 2016) and cross-sectional, which limits its applicability to the dynamics of the labor market today and makes it impossible to evaluate occupational stability or long-term career satisfaction [37]. Second, the model only takes into account self-reported job preferences, which could not accurately reflect real employment results. Third, only RIASEC interest ratings are included in the feature set,

leaving out crucial cognitive and psychometric characteristics like motivation and personality that have been demonstrated to increase the accuracy of career predictions [1], [16]. Furthermore, the existing system lacks real-time adaptation and user input integration, making it static and non-interactive. By using longitudinal tracking of people's actual career pathways to assess real-world effect, future research should fill in these gaps. Personalization and predictive power may be improved by including psychological factors, such as motivation models like Self-Determination Theory and the Big Five personality characteristics (e.g., BFI-10) [13]. Additionally, as proposed by Hossain et al., developing an adaptive, web-based decision support system (DSS) with real-time feedback and iterative learning capabilities might greatly enhance user engagement and result alignment [23]. Finally, the system's usefulness would be expanded by extending the framework to assist mid-career professionals who are considering transitions or reskilling possibilities. To guarantee equal guidance across user subgroups, formal fairness and bias audits utilizing demographic-aware measures and debiasing algorithms should also be given top priority.

The suggested framework not only increased predictive accuracy but also showed improved robustness and stability, as evidenced by consistent performance across cross-validation folds. The machine learning models were able to capture non-linear relationships between occupational categories and vocational interest profiles that are not well modeled by conventional rule-based approaches by utilizing extensive psychometric data. This work's emphasis on interpretability, which keeps predictions clear and useful for practical career counseling in higher education settings, is one of its main contributions. The framework is intended as a decision-support tool that enhances counselors' expertise rather than taking the place of human judgment. The outcomes also spur the system's development into an adaptive, web-based decision support platform with real-time feedback, allowing for more individualized and responsive career counseling.

Psychometric data driven student counselling system is improved in higher education comparing with the traditional approaches. Future research should focus on expanding data sources and exploring sophisticated architectures to improve model sophistication and dataset diversity. Personalized career recommendations could be offered by creating interactive career recommendation systems using machine learning models. This work can be expanded in a number of significant ways by future research. To determine whether anticipated RIASEC-aligned career paths result in long-term job stability, job satisfaction, and career advancement, longitudinal career tracking should first be included. Prediction accuracy and personalization could be further improved by incorporating additional psychometric constructs, such as motivation models (e.g., Self-Determination Theory), cognitive abilities, and personality traits (e.g., Big Five/BFI-10). Third, the framework can be extended to interactive and adaptive decision support systems, where real-time career recommendations are updated based on user feedback. In order to guarantee equitable recommendations across gender, geography, and socioeconomic groups, future research should also investigate the integration of system-level deployment and scalability assessment, reinforcement learning with real-time user feedback and longitudinal career monitoring for fairness-aware and bias-mitigation strategies. From a methodological standpoint, hybrid architectures that combine explainable AI (XAI) methods (like SHAP-based reasoning) with interpretable machine learning could increase adoption and trust in counseling settings. Lastly, expanding the model to assist mid-career professionals and reskilling pathways would increase the framework's applicability outside of student populations. However, the study's limitations include the quality and size of the dataset, potential biases, and the need for incorporating additional criteria for career alignment.

## AUTHOR'S CONTRIBUTIONS

Minakshi Roy: Conceptualization, methodology, formal analysis, writing–original draft, validation. Kalpana Sharma: Supervision, formal analysis, validation, writing–review and editing.

## CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

## DATA AVAILABILITY

The data presented in this study are available accessible via the Open Science Framework project website.

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